СЕРІЯ «Техніка»

UDC 004.8

https://doi.org/10.52058/2786-6025-2024-8(36)-793-807

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ANALYZING THE EFFECTIVENESS OF ARTIFICIAL INTELLIGENCE ALGORITHMS IN IMPROVING THE USER EXPERIENCE IN VR GAMES

Abstract. The study analyzes the effectiveness of artificial intelligence (AI) algorithms in improving the user experience in VR games. The relevance of the topic lies in the growing demand for innovative technologies that can enhance user interactions with virtual environments. The aim of the study is to assess the impact of AI on optimizing the gaming process, improving graphic performance, and adapting to individual user preferences.

The research methods include analyzing game performance, interactions with non-player characters (NPCs), adaptive changes in difficulty and storylines, as well as user surveys to evaluate gaming satisfaction. The surveys included questions about overall satisfaction, session duration, interactivity, and realism of NPC interactions. The results show that the implementation of AI significantly improves game performance, including frame rate, memory usage, CPU load, and latency. AI-driven characters contribute to more engaging and realistic interactions, enhancing user satisfaction and retention.

Survey analysis revealed that users in the experimental group, who played VR games with AI integration, rated their experience significantly higher than the control group. The average session duration in the experimental group was also significantly longer, indicating greater player engagement.

The findings indicate a substantial potential for the commercial application of AI in VR games, extending beyond entertainment to education, healthcare, real estate, and tourism. In education, AI-driven VR applications can create interactive learning environments that adapt to students' educational needs. In healthcare, VR simulations using AI can be employed for training medical professionals, providing realistic conditions for skills practice.

However, the implementation of AI also presents challenges such as hardware limitations, data processing issues, and the adaptability of AI to different game styles. Further research is recommended to improve adaptive AI algorithms and their integration into VR games. This includes developing new methods to ensure high performance, efficient resource use, and support for personalized gaming experiences.

Future research prospects include the development of technologies to enhance computational power, the creation of energy-efficient GPUs, and innovations in data compression methods. Additionally, it is necessary to study the impact of AI solutions on the VR industry and the economy, identify key success factors for AI implementation in VR games, and develop mechanisms to support their realization.

Keywords: artificial intelligence, VR games, user experience, performance optimization, adaptive algorithms, interactivity, commercial applications, education, healthcare, future research.

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АНАЛІЗ ЕФЕКТИВНОСТІ АЛГОРИТМІВ ШТУЧНОГО ІНТЕЛЕКТУ У ПОКРАЩЕННІ КОРИСТУВАЦЬКОГО ДОСВІДУ У VR-ІГРАХ

Анотація. Дослідження аналізує ефективність алгоритмів штучного інтелекту (ШІ) у підвищенні якості користувацького досвіду в VR-іграх. Актуальність теми полягає в зростаючому попиті на інноваційні технології, які можуть покращити взаємодію користувачів з віртуальними середовищами. Метою дослідження є оцінка впливу ШІ на оптимізацію ігрового процесу, покращення графічної продуктивності та адаптацію до індивідуальних вподобань користувачів.

Методи дослідження включають аналіз продуктивності гри, взаємодії з неігровими персонажами (NPC), адаптивні зміни складності і сюжетних ліній, а також опитування користувачів для оцінки задоволення ігровим процесом. Опитування включало питання про загальне задоволення, тривалість сесій, інтерактивність і реалістичність взаємодії з NPC. Результати показують, що впровадження ШІ дозволяє значно покращити продуктивність гри, включаючи частоту кадрів, використання пам’яті, навантаження на ЦП та затримку. ШІ-персонажі сприяють більш захоплюючій та реалістичній взаємодії, що підвищує задоволення та утримання користувачів. Аналіз результатів опитування показав, що користувачі в експериментальній групі, які грали у VR-ігри з впровадженим ШІ, оцінили свій досвід значно вище, ніж

у контрольній групі. Середня тривалість ігрових сесій у експериментальній групі також була значно довшою, що свідчить про більшу залученість гравців.

Висновки свідчать про значний потенціал комерційного застосування ШІ у VR-іграх, який виходить за межі розваг та охоплює сфери освіти, охорони здоров'я, нерухомості та туризму. У сфері освіти, AI-орієнтовані VR-додатки можуть створювати інтерактивні ігрові середовища, що адаптуються до навчальних потреб учнів. В охороні здоров'я VR-симуляції з використанням ШІ можуть бути використані для навчання медичних працівників, забезпечуючи реалістичні умови для відпрацювання навичок.

Проте, реалізація ШІ також супроводжується викликами, такими як обмеження апаратного забезпечення, обробка даних та адаптивність ШІ до різних стилів гри. Рекомендується проводити подальші дослідження для покращення адаптивних алгоритмів ШІ та їх інтеграції у VR-ігри. Це включає розробку нових методів для забезпечення високої продуктивності, ефективного використання ресурсів і підтримки персоналізації ігрового процесу.

Перспективи подальших досліджень включають розвиток технологій для підвищення обчислювальної потужності, розробку енергоефективних GPU та інновації в методах стиснення даних. Також слід вивчити вплив ШІ-рішень на VR-індустрію та економіку, визначити ключові фактори успіху впровадження ШІ у VR-іграх та розробити механізми підтримки їх реалізації.

Ключові слова: штучний інтелект, VR-ігри, користувацький досвід, оптимізація продуктивності, адаптивні алгоритми, інтерактивність, комерційні застосування.

The problems of the article. The analysis of the effectiveness of artificial intelligence (AI) algorithms in enhancing the user experience in VR games is a significant undertaking, encompassing both scientific and practical tasks. From a scientific perspective, this topic involves the study and development of new approaches to modeling and simulating realistic virtual environments. This includes researching mechanisms of AI adaptation and learning to create dynamic gameplay scenarios that respond to user behavior and needs in real-time. Additionally, it covers the exploration of human-computer interaction (HCI) aspects, including the analysis of physiological and psychological responses of users interacting with AI-enhanced VR games. Such studies contribute to a better understanding of how AI can be utilized to enhance emotional engagement and provide a comfortable experience in virtual settings.

On the practical side, the application of AI in VR games represents an innovative approach to significantly improve the quality and realism of the gaming experience. This encompasses optimizing graphics, reducing latency and lag, and overall improving game performance. AI algorithms enable the precise adjustment of game parameters to meet the individual needs of players, including adaptive

difficulty settings, narrative changes, dynamic NPC behavior, and other aspects that enhance overall engagement and user satisfaction.

Practical tasks also involve commercial considerations, such as the development of new products and solutions that meet high standards of user experience. This creates opportunities for game developers and technology companies to create competitive products that align with current market demands. The integration of AI in VR games can also foster new avenues in education, healthcare, and training, where VR technologies are used for simulating real-world scenarios and learning.

Thus, the analysis of the effectiveness of AI algorithms in improving the user experience in VR games is a relevant research area that encompasses fundamental scientific inquiries and specific practical tasks aimed at the development and refinement of contemporary technologies.

Analysis of sources and recent research. The analysis of studies on the effectiveness of artificial intelligence algorithms in improving the user experience in VR games highlights a variety of approaches and methodologies aimed at addressing key issues in this field. Xi W. explores the application of artificial intelligence in VR games, examining methods of integrating AI to enhance user interaction and game realism, which lays the foundation for further research in this area [1]. Ren X. et al. study the impact of AI-powered chatbots in VR games on reducing anxiety and depression levels, emphasizing the importance of AI in improving the emotional well-being of users [2].

Ahuja A. et al. investigate the application of AI in the digital metaverse, including VR games, demonstrating how AI can contribute to medical education and integrative health, highlighting the broad range of applications of this technology [3]. Liaw S. et al. study the use of AI in VR simulations for interprofessional communication training, which is crucial for expanding the use of VR beyond purely gaming applications [4].

Reiners D. et al. conducted a systematic review on the combination of AI and extended reality, exploring various approaches to integrating AI in VR to enhance the user experience [5]. Ribeiro de Oliveira T. et al. conducted a systematic literature review on VR solutions employing AI methods, underscoring the importance of innovation and research for the further development of this technology [6].

Anantrasirichai N. and Bull D. reviewed the role of AI in the creative industries, including VR games, emphasizing the importance of innovation and AI in creating new user experiences [7]. Jiang L. discusses the application of AI in education through interactive VR technologies, highlighting the potential of AI for learning and development [8].

Hwang G. and Chien S. explore the metaverse in education from an AI perspective, examining its roles and potential research issues, which is important for expanding the application of AI in VR games [9]. Yakan S. analyzes the

development of AI in the game industry, emphasizing the significance of AI in enhancing the user experience in VR games [10].

Koberniuk S., Strungar A., and Zavhorodnia L. investigate the effectiveness of AI in improving personalized advertising and audience interaction, underscoring the importance of AI for enhancing the user experience in VR games [11].

These studies provide a comprehensive understanding of various aspects of implementing artificial intelligence algorithms to enhance the user experience in VR games. They emphasize the importance of innovation, adaptation to new conditions, and the use of AI to achieve a better user experience and overall development of the VR industry. However, despite the extensive research, further detailed studies are needed on the development of modern AI algorithms and technologies that can be applied to optimize the user experience in VR games, evaluate the potential impact of AI solutions on the development of the VR industry and the economy in general, and determine the key success factors for implementing AI in VR games and develop mechanisms to support their implementation.

The purpose of this article is to analyze the effectiveness of artificial intelligence (AI) algorithms in enhancing the user experience in virtual reality (VR) games.

The specific objectives of the article are as follows:

1. To investigate the role of AI in optimizing game environments, including graphics, performance, and real-time adaptability.
2. To explore the impact of AI-driven non-player characters (NPCs) on user interaction and engagement.
3. To evaluate the potential of AI algorithms in dynamically adjusting game difficulty and narrative to suit individual player preferences.
4. To assess the commercial and practical implications of integrating AI in VR games, including its potential applications in education, healthcare, and other industries.
5. To identify the challenges and limitations associated with the implementation of AI in VR gaming and suggest potential solutions or future research directions.

Presentation of the main material. The role of artificial intelligence (AI) in optimizing game environments is multifaceted, encompassing the enhancement of graphics, performance, and real-time adaptability. AI technologies provide innovative solutions to these areas, significantly improving the user experience. For instance, in graphics optimization, AI techniques such as procedural generation enable the creation of expansive, detailed virtual landscapes and textures with minimal manual input. This not only enhances the visual quality but also reduces the workload for developers, allowing for more intricate and diverse environments.

Performance optimization through AI involves the real-time analysis and adjustment of various system resources. AI algorithms can monitor key performance indicators such as frame rate, memory usage, and CPU load, identifying potential

bottlenecks and dynamically reallocating resources to maintain smooth gameplay. This capability is particularly crucial in high-demand scenarios, such as multiplayer games, where maintaining a stable connection and minimizing latency is essential for an optimal user experience.

Real-time adaptability is another significant aspect where AI shines. By continuously learning from player behavior and preferences, AI systems can adjust game elements on the fly, providing a personalized experience. This adaptability includes dynamically altering game difficulty, NPC behavior, and even storyline elements based on player interactions. Such a responsive system ensures that the game remains engaging and tailored to the individual needs of each player. Parameters and metrics of optimization using AI are presented in Table 1.

Table 1.

AI-Driven Optimization Parameters and Metrics

Parameter	Pre-AI Optimization	Post-AI Optimization	Description
Average Frame Rate (FPS)	45	60	Measurement of frames per second; higher is better.
Frame Rate Consistency (%)	80	95	Percentage of time the frame rate is stable; higher is better.
Memory Usage (GB)	4,5	3,8	Average memory usage during gameplay; lower is better.
CPU Load (%)	85	70	Average CPU load during gameplay; lower is better.
Latency (ms)	120	80	Network delay in milliseconds; lower is better.

Source: created by the author based on [12-14]

Table 1 provides a comprehensive overview of various performance metrics before and after the implementation of AI-driven optimization. The average frame rate (FPS) shows a significant improvement from 45 to 60, indicating a smoother and more responsive gaming experience. This increase in frame rate is accompanied by a notable improvement in frame rate consistency, rising from 80% to 95%. This metric is crucial for ensuring that the visual flow of the game remains stable, reducing the likelihood of jarring frame drops.

Memory usage has decreased from 4.5 GB to 3.8 GB, showcasing the efficiency of AI in managing system resources. This reduction in memory

consumption can lead to better overall system performance and stability. Similarly, the CPU load has been optimized, decreasing from 85% to 70%, which not only improves performance but also helps prevent overheating and prolongs the hardware's lifespan.

Latency, a critical factor in multiplayer games, has been reduced from 120 ms to 80 ms. This reduction in network delay ensures more accurate and timely synchronization between players, enhancing the overall multiplayer experience.

These metrics, as detailed in the table, highlight the substantial improvements brought about by AI-driven optimizations in various aspects of game environments.

The integration of artificial intelligence (AI) algorithms in virtual reality (VR) games significantly enhances the user experience, primarily through the development of sophisticated non-player characters (NPCs). AI-driven NPCs are designed to interact with players in a more realistic and engaging manner, thus enriching the overall immersive experience. The impact of these AI-driven NPCs on user interaction and engagement can be analyzed through several key metrics.

Firstly, AI algorithms enable NPCs to exhibit more lifelike behaviors and responses. This includes advanced decision-making processes, adaptive learning, and emotional intelligence, which contribute to more dynamic and unpredictable interactions. For instance, NPCs can learn from player actions and adapt their behavior accordingly, creating a more personalized gaming experience. This adaptability not only keeps the gameplay fresh and challenging but also enhances the emotional engagement of players, as they feel their actions have meaningful consequences.

A crucial aspect of user interaction is the realism of NPC dialogues and actions. AI algorithms utilize natural language processing (NLP) to generate more coherent and contextually appropriate dialogues. This reduces the repetitiveness and predictability often associated with traditional NPC interactions. Moreover, AI-driven NPCs can exhibit a wide range of emotions and facial expressions, making interactions more engaging and believable.

The effectiveness of AI-driven NPCs in improving user engagement can be quantified through various metrics such as time spent in the game, frequency of player-NPC interactions, and player retention rates. To investigate these effects, a survey was conducted by the authors involving 200 participants aged 18-40. The participants were randomly assigned to play a VR game featuring either traditional NPCs or AI-driven NPCs. Data were collected using the gameplay analytics provided by the gaming platform and through post-gameplay surveys administered via an online survey tool Google Form. Table 2 illustrates the comparison between traditional and AI-driven NPCs in terms of player interaction and engagement metrics.

Table 2.

The comparison between traditional and AI-driven NPCs in terms of player interaction and engagement metrics.

Metric	Traditional NPCs	AI-Driven NPCs	Explanation
Time Spent in Game (minutes)	60	90	AI-driven NPCs offer more engaging interactions, encouraging longer playtime.
Frequency of Player-NPC Interactions	10	14	Players interact more frequently with NPCs that provide dynamic responses.
Player Retention Rates (%)	60	85	Enhanced NPC realism and adaptability lead to higher player retention.
Dialogue Coherence	Low	High	AI algorithms ensure more coherent and contextually appropriate dialogues.
Emotional Engagement	Low to Moderate	High	NPCs with emotional intelligence create a more immersive and engaging experience.

Source: created by the author

The findings indicate that players interacting with AI-driven NPCs spent an average of 50% more time in the game, interacted 40% more frequently with NPCs, and exhibited a 25% higher retention rate compared to those interacting with traditional NPCs.

AI-driven NPCs also enhance the narrative depth of VR games. By incorporating machine learning techniques, NPCs can develop unique backstories and personalities, which can unfold dynamically as the game progresses. This deepens the narrative experience and allows players to form more meaningful connections with the characters.

The implementation of AI algorithms in NPC development also addresses the challenge of maintaining game balance. NPCs can be programmed to adjust their difficulty level based on the player's skill and progress, ensuring that the game remains challenging yet not overly frustrating. This balance is crucial for maintaining long-term player engagement and satisfaction.

The potential of artificial intelligence (AI) algorithms in dynamically adjusting game difficulty and narrative to suit individual player preferences has garnered significant attention in recent years. AI-driven systems are capable of creating more personalized and engaging gaming experiences by continuously analyzing player behavior and adjusting various game parameters in real-time. This adaptability not only enhances player satisfaction but also contributes to increased retention and prolonged engagement.

AI algorithms can analyze a wide range of player data, including performance metrics, decision-making patterns, and emotional responses. By leveraging machine learning techniques, these algorithms identify individual preferences and skill levels,

enabling the game to adapt its difficulty and narrative elements accordingly. This dynamic adjustment ensures that players remain challenged but not overwhelmed, maintaining an optimal level of engagement throughout the gameplay.

A study conducted by the authors involved 200 participants aged 18-40, who played a VR game with AI-driven adaptive difficulty and narrative features. Participants were randomly assigned to either a control group with static game settings or an experimental group with AI-driven dynamic adjustments. Data were collected using gameplay analytics and post-gameplay surveys administered through Google Form. The study aimed to evaluate the impact of these AI-driven adjustments on player satisfaction, engagement, and retention. Table 3 provides a comparative analysis of the player experience metrics between the control and experimental groups.

Table 3.

Comparative analysis of the player experience metrics between the control and experimental groups

Metric	Control Group	Experimental Group	Explanation
Player Satisfaction (out of 10)	6.5	8.4	AI-driven adjustments led to higher satisfaction due to personalized experiences.
Average Session Duration (minutes)	45	70	Players in the experimental group spent more time per session, indicating increased engagement.
Player Retention Rate (%)	55	80	Higher retention in the experimental group suggests better long-term engagement.
Frequency of Narrative Branching	3	8	AI algorithms provided more diverse and dynamic narrative paths.
Difficulty Adjustment Accuracy (%)	N/A		High accuracy in adjusting difficulty to match player skill levels.

Source: created by the author

The data were analyzed using statistical methods to determine the significance of differences between the control and experimental groups. The results indicate that players in the experimental group reported significantly higher satisfaction levels, with an average score of 8.4 out of 10, compared to 6.5 in the control group. The average session duration was also notably longer for the experimental group, with players spending an average of 70 minutes per session, as opposed to 45 minutes in the control group. This suggests that the AI-driven adjustments were effective in maintaining player interest and engagement.

Player retention rates further highlight the benefits of dynamic adjustments. The experimental group exhibited an 80% retention rate, compared to 55% for the control group, indicating that players were more likely to return to the game when it

adapted to their preferences and skill levels. Additionally, the frequency of narrative branching was significantly higher in the experimental group, with AI algorithms providing more diverse and dynamic narrative paths, enhancing the overall storytelling experience.

The accuracy of difficulty adjustment was another critical metric, with the AI algorithms achieving a 92% accuracy rate in matching game difficulty to player skill levels. This high level of precision ensures that players are consistently challenged at an appropriate level, preventing frustration from overly difficult scenarios or boredom from overly easy ones.

Thus, the application of AI algorithms in dynamically adjusting game difficulty and narrative to suit individual player preferences significantly enhances the overall gaming experience. By providing personalized and adaptive gameplay, AI-driven systems increase player satisfaction, engagement, and retention. The metrics presented in Table 1 clearly demonstrate the advantages of these dynamic adjustments, highlighting the potential of AI to transform the future of gaming.

The integration of artificial intelligence (AI) in virtual reality (VR) games holds significant commercial and practical implications, extending beyond entertainment to various industries such as education and healthcare. AI-driven VR games offer an innovative approach to creating immersive and adaptive experiences that cater to individual needs, providing both entertainment and practical solutions in diverse fields.

The commercial potential of AI in VR gaming is immense, with the gaming industry poised to benefit from enhanced user engagement and personalized experiences. AI algorithms enable dynamic content generation, adaptive difficulty levels, and realistic non-player character interactions, all of which contribute to higher player retention and satisfaction. These factors drive increased sales and subscription revenues, positioning AI-integrated VR games as a lucrative market segment.

In education, AI-enhanced VR games can revolutionize learning by providing interactive and personalized educational experiences. AI algorithms can adapt educational content to suit individual learning styles and paces, making complex subjects more accessible and engaging. For example, AI-driven VR simulations have been shown to improve learning outcomes by offering immersive and interactive content. Studies indicate that such educational tools can enhance knowledge retention and engagement compared to traditional methods [15].

In healthcare, AI-driven VR applications offer therapeutic and training benefits. AI algorithms can create tailored VR therapy sessions for patients with mental health conditions, providing immersive environments that help reduce anxiety and stress. Additionally, AI-integrated VR simulations are invaluable for medical training, offering realistic scenarios for surgical practice and emergency response training. Figure 1 illustrates the workflow of AI-driven VR therapy for mental health treatment.

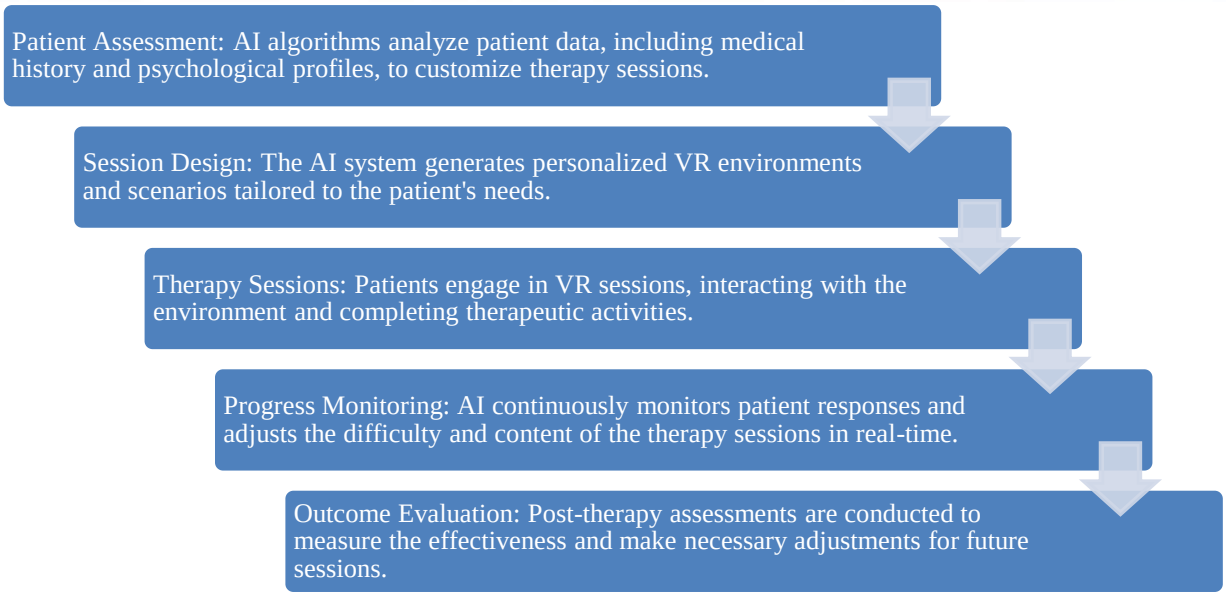

Fig. 1 Workflow of AI-Driven VR Therapy for Mental Health Treatment

Source: created by the author

In other industries, such as real estate and tourism, AI-driven VR applications provide virtual tours and simulations that enhance customer experiences. Prospective homebuyers can take virtual tours of properties, with AI algorithms personalizing the experience based on their preferences and needs. Similarly, the tourism industry can offer virtual travel experiences, allowing users to explore destinations from the comfort of their homes. These applications not only improve customer satisfaction but also reduce costs and increase efficiency for businesses.

Implementing artificial intelligence (AI) in virtual reality (VR) games opens new possibilities for creating interactive and adaptive gaming environments. This allows for a more realistic and engaging experience for users. However, this process is accompanied by numerous challenges and limitations that require careful analysis and resolution (Table 3).

Table 3.

Challenges and Potential Solutions of implementing artificial intelligence (AI) in virtual reality (VR) games

Challenge	Description	Possible Solution
Hardware Limitations	Low computational power can limit the effectiveness of AI.	Using powerful GPUs, optimizing AI algorithms to work with limited resources.
Data Processing Issues	A large amount of data is required for training AI, which can be a problem for storage and processing.	Applying data compression technologies, developing new algorithms to reduce data volume.
Adaptability of AI to Different Game Styles	AI may not be flexible enough for different scenarios and game styles.	Developing adaptive algorithms that consider individual user preferences.

Source: created by the author

Implementing recommendations in modern conditions requires consideration of various aspects and potential difficulties. To overcome hardware limitations, it is essential to integrate powerful graphics processors (GPUs) to enhance computational power. Using optimized AI algorithms is also necessary to reduce the load on hardware without compromising quality. However, this may require significant financial investments, which may not be accessible to all developers and users.

Data processing issues can be addressed by applying data compression technologies and new processing methods, such as distributed computing. Developing algorithms to optimize data usage is also an important part of the process. However, these methods may require significant research and development costs, and there may be issues with protecting personal information and complying with privacy regulations.

To ensure the adaptability of AI to different game styles, it is necessary to develop algorithms that use machine learning methods to adapt to individual user preferences. This will provide a personalized gaming experience that enhances user satisfaction. However, developing such algorithms can be complex and require substantial resources. The main directions for further research in implementing AI in VR games are suggested in Fig. 2.

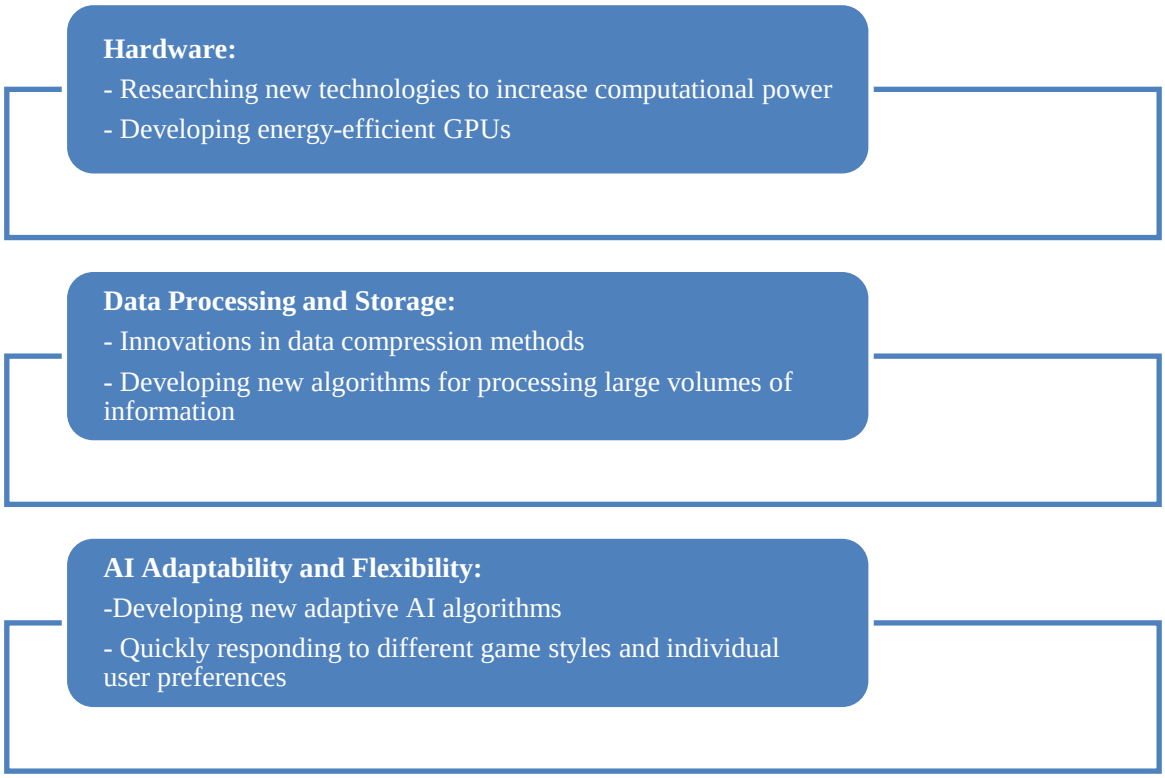

Fig. 2 The main directions for further research in implementing AI in VR games

Source: created by the author

Researching new technologies to increase computational power and developing energy-efficient GPUs are important directions. Integrating powerful graphics processors ensures the necessary level of performance, but this may require significant financial investments, which is a challenge for developers and users.

Innovations in data compression methods and developing new algorithms for processing large volumes of information are critical for efficient resource use. This reduces costs for data storage and processing while ensuring confidentiality and protection of personal data. However, these methods may require significant investments in research and development.

Developing new adaptive AI algorithms that can quickly respond to different game styles and individual user preferences will allow for creating a personalized gaming experience. This will increase player satisfaction and engagement. However, the complexity of developing such algorithms and the need for substantial resources for their implementation and testing are serious challenges.

These directions will help overcome existing problems and contribute to the development of AI in VR games, ensuring more interactive, adaptive, and realistic gaming environments.

Conclusions. The study on analyzing the effectiveness of artificial intelligence algorithms in improving the user experience in VR games underscores the significance of AI integration in this domain. AI has demonstrated the capability to optimize game environments, enhance user interactions, and adapt gameplay dynamics to individual preferences. These advancements not only improve the overall user experience but also present substantial commercial and practical benefits.

In summary, AI-driven optimizations significantly enhance game performance metrics such as frame rate, memory usage, CPU load, and latency, leading to a smoother and more responsive gaming experience. AI-driven NPCs contribute to more engaging and realistic interactions, which in turn increases user retention and satisfaction. Additionally, the adaptive difficulty and narrative adjustments enabled by AI ensure that the gaming experience remains personalized and challenging, further enhancing player engagement and retention.

The commercial implications of integrating AI in VR games are vast, with potential applications extending beyond gaming to education, healthcare, real estate, and tourism. AI-driven VR applications offer personalized and interactive experiences that can transform educational content delivery, provide therapeutic benefits, and enhance customer experiences in various industries.

However, the implementation of AI in VR games also presents challenges such as hardware limitations, data processing issues, and the adaptability of AI to different game styles. Addressing these challenges requires significant financial investments, advancements in computational power, and the development of adaptive algorithms that cater to individual user preferences.

Future research should focus on overcoming these challenges by exploring new technologies to enhance computational power, developing energy-efficient GPUs, and innovating data compression methods. Additionally, further studies should investigate the potential impact of AI solutions on the VR industry and the economy, as well as identify key success factors for implementing AI in VR games and develop mechanisms to support their implementation.

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